

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #2: due Thursday, Jan. 29th, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

1. In Pharloom, Hornet encounters different types of enemies with the following probabilities: Pilgrims (40%), Ants (35%) and Flies (25%). When defeated, each enemy might drop Geo (a form of currency) as follows:
 - Pilgrims drop Geo with probability 0.6
 - Ants drop Geo with probability 0.3
 - Flies drop Geo with probability 0.8
 - (a) (10 points.) What is the overall probability that a defeated enemy drops Geo? What probability concept are you using?
 - (b) (10 points.) Given that Geo dropped, what is the probability the enemy was an Ant? What probability concept are you using?
 - (c) (10 points.) If we know that Geo dropped, does that make it more or less likely for the enemy to be a Fly (vs. not knowing that information)? What are the two probabilities that are involved called?
2. Jason and Stevie are each scored independently by the Taskmaster on a scale of 0 to 5. That is, each of them gets one of the scores $\{0, 1, 2, 3, 4, 5\}$ with equal probability $\frac{1}{6}$. Let Jason's score be J and Stevie's score be S .
 - (a) (10 points.) Compute $P(J + S \geq 8 \mid S \text{ is odd})$. Leave your answer as a simple fraction.
 - (b) (10 points.) Are the events $\{J + S \text{ is odd}\}$ and $\{J \text{ is odd}\}$ independent? Justify your answer.
 - (c) (10 points.) Are the events $\{J = 2\}$ and $\{J + S \geq 5\}$ independent? Justify your answer.
3. Suppose that in Balatro, each time a player enters a shop, the shop (independently of other shops) contains at least one Rare Joker with probability $p = 0.4$. Compute each of the following values to 3 decimal places.
 - (a) (10 points.) The probability that exactly 3 out of the next 8 shops contain a Rare Joker.
 - (b) (10 points.) The probability that at least 3 of the next 6 shops contain a Rare Joker.
 - (c) (10 points.) The probability that the first shop containing a Rare Joker is the 5th shop visited.
 - (d) (10 points.) A lower bound on the probability that in a tournament with 12000 total shop visits, between 4600 and 5000 shops contain a Rare Joker (use Bernoulli's theorem).